

ADVANCED SYSTEMS AND ALGORITHMS

Section 1 - Algorithms and Data

Candidate instructions

Answer in the printed order only when convenient; every labelled sub-part is a separate question. Show intermediate reasoning for calculations. If a diagram is requested, label all edges or axes. There are 20 marks on this fixture; some questions deliberately contain multi-line data.

SECTION A - CORE REASONING

1. [3]

Define amortized analysis. Use the potential method to analyse a dynamic table that doubles when full and shrinks when one quarter full.

2 (a). [4]

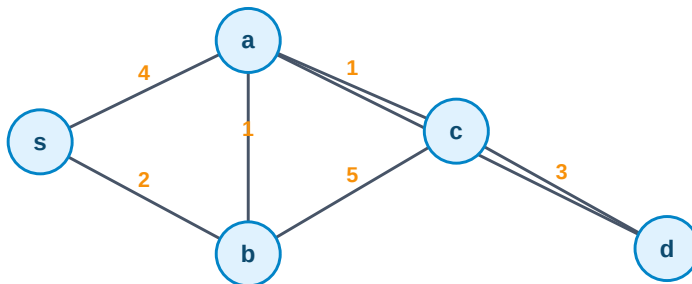
Solve the recurrence $T(n) = 2T(n/2) + n \log n$ using a suitable form of the Master theorem. State any regularity assumption you use.

2 (b). [2]

Give a tight asymptotic bound for the total work if the recurrence is evaluated for every power-of-two input from 1 through n .

3. [4]

For the directed weighted graph below, run Dijkstra's algorithm from source s . Report the final distance vector in the order (s, a, b, c, d) and the predecessor of each non-source vertex.



4 (a). [3]

Explain the CAP theorem using a network partition between two replicas. Name one design choice that favours availability and one that favours consistency.

4 (b). [3]

Compare serializable isolation with snapshot isolation. Give one write-skew example that snapshot isolation may permit.

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Section 2 - Theory, ML and Security

SECTION B - STRUCTURED ANSWERS

5. [4]
Given Orders(order_id, customer_id, amount, created_at) and Customers(customer_id, region), write a SQL query that returns the top two customers per region by total order amount in the last 30 days. Include ties.
Use a window function or explain an equivalent approach.
- 6 (a). [4]
Prove that $L = \{ 0^n 1^n \mid n \geq 0 \}$ is not regular using the pumping lemma. State clearly where the contradiction occurs.
- 6 (b) (i). [2]
Give a PDA transition sketch for the language in 6 (a). A compact labelled diagram is sufficient.
- 6 (b) (ii). [2]
State whether the grammar $S \rightarrow 0S1 \mid \epsilon$ is ambiguous. Justify your answer in one or two sentences.
- 7 (a). [3]
For one gradient step of logistic regression, write the vector update for weights w and bias b , using learning rate η and binary cross-entropy loss.
- 7 (b). [3]
A fraud detector changes its threshold from 0.50 to 0.20. Predict the direction of change for precision, recall and the false-positive rate, and explain why.
8. [4]
Insert the keys 41, 38, 31, 12, 19, 8 into an initially empty red-black tree. Draw the final tree and mark the colour of every node.
9. [3]
Describe a two-phase commit timeline in which the coordinator crashes after sending PREPARE but before sending COMMIT. Explain what each participant can safely do.
10. [3]
Review this pseudocode: ``if user.is_admin(): file.write(request.body); log(request.user);`` Identify two security problems and give one mitigation for each.

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Section 3 - Design and Quantitative Reasoning

SECTION C - APPLICATION**11 (a).****[3]**

Design an idempotent POST /payments endpoint. List the minimum request fields, the idempotency-key behaviour, and the response returned when a key is replayed.

11 (b).**[2]**

Give one reason a database transaction alone does not guarantee exactly-once payment effects when a downstream provider is involved.

12.**[5]**

A classifier is evaluated on 2,000 cases. The confusion matrix is TP = 168, FP = 42, FN = 32, TN = 1,758. Calculate accuracy, precision, recall and F1 score to three decimal places.

Reference data block

Metric	Positive	Negative	Total
Actual	200	1,800	2,000
Predicted	210	1,790	2,000

Use the confusion-matrix values in the question, not the rounded reference block, for your calculations.